TRAINITY PROJECT 6

Bank Loan Case Study

<u>Project Description:</u> This project involves **Exploratory Data Analysis (EDA)** to identify patterns in loan applications and predict **loan defaults**. The company faces two key risks: **losing business** by rejecting capable applicants and **financial loss** from approving defaulters. The analysis focuses on customer and loan attributes to determine key factors influencing default, helping optimize loan approvals, reduce risks, and adjust lending terms.

<u>TASKS:</u>

A. **Identify Missing Data and Deal with it Appropriately:** As a data analyst, you come across missing data in the loan application dataset. It is essential to handle missing data effectively to ensure the accuracy of the analysis.

- **Task:** Identify the missing data in the dataset and decide on an appropriate method to deal with it using Excel built-in functions and features.
- **Hint:** Utilize Excel functions like COUNT, ISBLANK, and IF to identify missing data. Consider using functions like AVERAGE or MEDIAN for imputation or other appropriate methods available in Excel.
- **Graph suggestion:** Create a bar chart or column chart to visualize the proportion of missing values for each variable.

B. Identify Outliers in the Dataset: Outliers can significantly impact the analysis and distort the results. You need to identify outliers in the loan application dataset.

- **Task:** Detect and identify outliers in the dataset using Excel statistical functions and features, focusing on numerical variables.
- **Hint:** Utilize Excel functions like QUARTILE, IQR, and conditional formatting to identify potential outliers. Consider applying thresholds or business rules to determine if the outliers are valid data points or require further investigation.
- **Graph suggestion:** Create box plots or scatter plots to visualize the distribution of numerical variables and highlight the outliers.

C. Analyze Data Imbalance: Data imbalance can affect the accuracy of the analysis, especially for binary classification problems. Understanding the data distribution is crucial for building reliable models.

- **Task:** Determine if there is data imbalance in the loan application dataset and calculate the ratio of data imbalance using Excel functions.
- **Hint:** Utilize Excel functions like COUNTIF and SUM to calculate the proportions of each class. Compare the class frequencies to assess data imbalance.
- **Graph suggestion:** Create a pie chart or bar chart to visualize the distribution of the target variable and highlight the class imbalance.

D. Perform Univariate, Segmented Univariate, and Bivariate Analysis: To gain insights into the driving factors of loan default, it is important to conduct various analyses on consumer and loan attributes.

- **Task:** Perform univariate analysis to understand the distribution of individual variables, segmented univariate analysis to compare variable distributions for different scenarios, and bivariate analysis to explore relationships between variables and the target variable using Excel functions and features.
- **Hint:** Utilize Excel functions like COUNT, AVERAGE, MEDIAN, and statistical functions for descriptive analysis. Utilize Excel features like filters, sorting, and pivot tables for segmented and bivariate analysis.
- **Graph suggestion:** Create histograms, bar charts, or box plots to visualize the distributions of variables. Create stacked bar charts or grouped bar charts to compare variable distributions across different scenarios. Create scatter plots or heatmaps to visualize the relationships between variables and the target variable.

E. **Identify Top Correlations for Different Scenarios:** Understanding the correlation between variables and the target variable can provide insights into strong indicators of loan default.

- **Task:** Segment the dataset based on different scenarios (e.g., clients with payment difficulties and all other cases) and identify the top correlations for each segmented data using Excel functions.
- **Hint:** Utilize Excel functions like CORREL to calculate correlation coefficients between variables and the target variable within each segment. Rank the correlations to identify the top indicators of loan default for each scenario.
- **Graph suggestion:** Create correlation matrices or heatmaps to visualize the correlations between variables within each segment. Highlight the top correlated variables for each scenario using different colours or shading.

Task A: Identify the missing data in the dataset and decide on an appropriate method to deal with it using Excel built-in functions and features.

Task B: Detect and identify outliers in the dataset using Excel statistical functions and features, focusing on numerical variables.

- Multiple columns in the dataset were found to be insignificant for the analysis, so the following columns were removed from the dataset for a clearer understanding. The columns were found to be insignificant because majority of the rows were null in these columns, replacing or deleting these rows would lead to significant changes in the dataset.

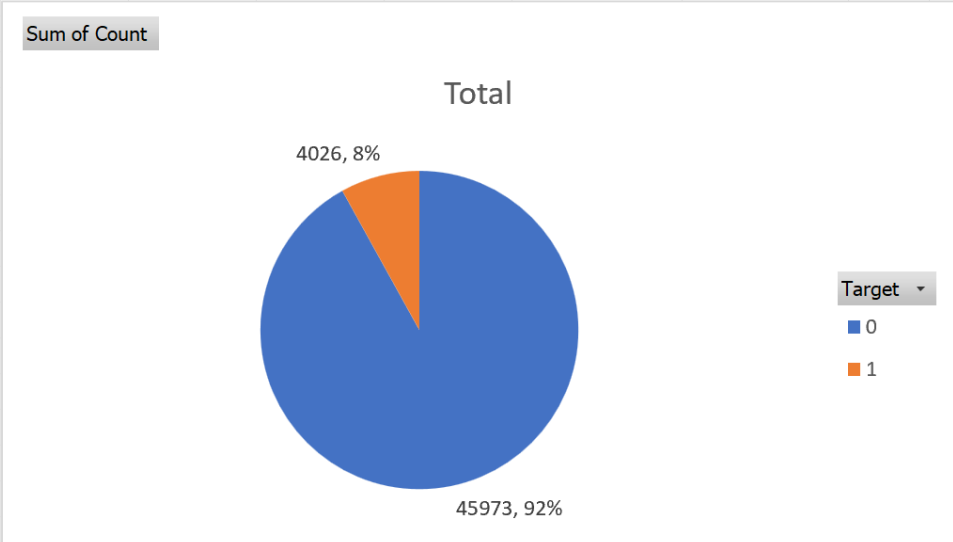
OWN_CAR_AGE	FLAG_MOBIL	FLAG_DOC_2
EXT_SOURCE_1	FLAG_WORK_PHONE	FLAG_DOC_3
APARTMENTS_AVG	FLAG_PHONE	FLAG_DOC_4
BASEMENTAREA_AVG	CNT_FAMILY_MEMBERS	FLAG_DOC_5
YEARS_BUILD_AVG	REGION_RATING_CLIENT_W_CITY	FLAG_DOC_6
COMMON_AREA_AVG	YEAR_BEGINEXPLUATATION_AVG	FLAG_DOC_7
ELEVATORS_AVG	YEAR_BEGINEXPLUATATION_MEDIAN	FLAG_DOC_8
ENTRANCES_AVG	EMERGENCYSTATE_MODE	FLAG_DOC_9
FLOORSMAX_AVG	HOUSE_TYPE_MODE	FLAG_DOC_10
FLOORSMIN_AVG	FONDKAPREMONT_MODE	FLAG_DOC_11
LANDAREA_AVG	FLAG_EMPLOY_PHONE	FLAG_DOC_12
LIVINGAPARTMENTS_AVG	FLAG_CONT_MOBILE	FLAG_DOC_13
LIVINGAREA_AVG	FLAG_EMAIL	FLAG_DOC_14
NONLIVINGAPARTMENTS_AVG	REGION_RATING_CLIENT	FLAG_DOC_15
NONLIVINGAREA_AVG	EXT_SOURCE_3	FLAG_DOC_16
APARTMENTS_MODE	YEAR_BEGINEXPLUATATION_MODE	FLAG_DOC_17
BASEMENTAREA_MODE	TOTAL_AREA_MODE	FLAG_DOC_18
YEARS_BUILD_MODE	DAYS_LAST_PHONE_CHANGE	FLAG_DOC_19
COMMON_AREA_MODE	WALLSMATERIAL_MODE	FLAG_DOC_20
ELEVATORS_MODE	FLAG_DOC_21	

- Changes were made in the following columns,
    - AMT_ANNUITY – one of the rows was null, replaced by the average value.
    - AMT_GOODS_PRICE – some rows were, which were replaced by the average value.
    - NAME_TYPE_SUITE – some rows were null, they were replaced by the highest occurring value in the column which was found using COUNTIF, the nulls were replaced by 'Unaccompanied'.
    - OCCUPATION_TYPE – some rows were null, replaced by the most occurring occupation which was 'Labourers'.
    - CNT_FAM_NUMBERS – one row was null replaced by the average number of family members of the customers.
    - ORGANIZATION_TYPE – 8924 rows were 'XNA' which were replaced by the highest occurring type 'Business Entity Type 3'.
    - CODE_GENDER – 2 rows had XNA as their gender which was replaced by the most occurring gender in their respective occupation types.
    - DAYS_EMPLOYED – There was an outlier present here with a value of 365243, 8924 rows of data had this number mentioned as their number of days of employment, this was replaced with the median days employed of '-1655'.
    - AMT_INCOME_TOTAL – There was an outlier present with a value '117000000', which was replaced by the median value.
- There were a few outliers present in some columns but those were neither deleted nor replaced as those columns were subjective which means they could be different for everyone therefore they were not touched.

Task C: Determine if there is data imbalance in the loan application dataset and calculate the ratio of data imbalance using Excel functions.

- Analysing the Target column:

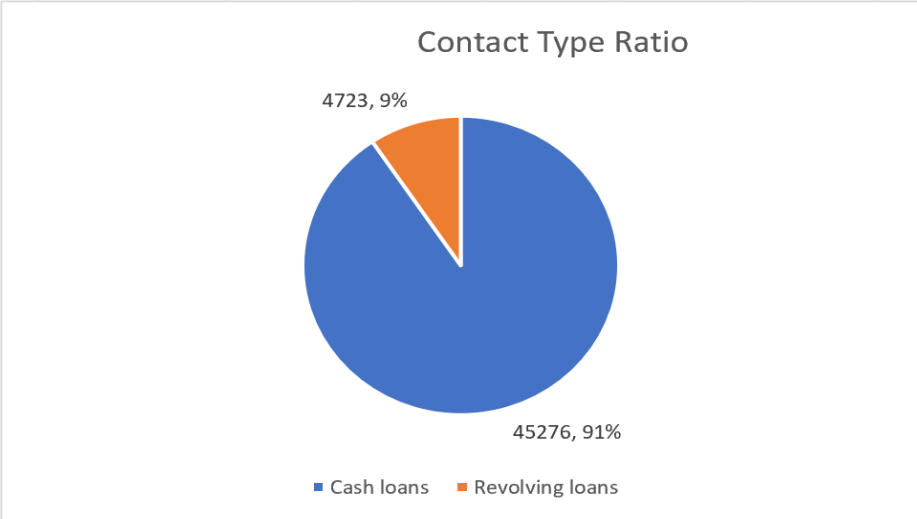
Target	Count	
0	45973	These people had no problems making their payment
1	4026	These people struggled making their payments



92% of the people had no issues making their payments where 8% people had problems while making payments.

- Analysing the Contract type column:

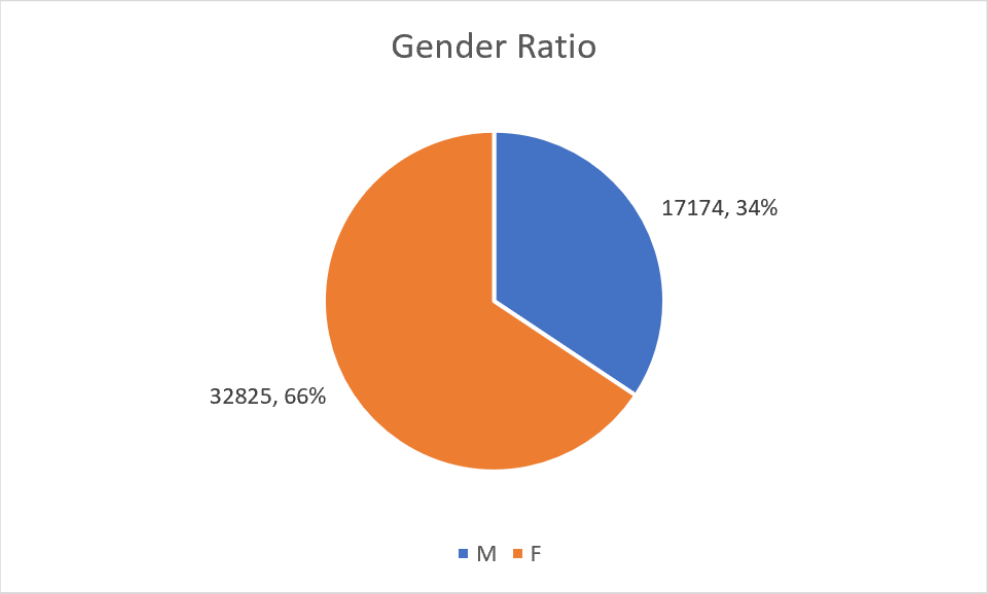
Type	Count
Cash loans	45276
Revolving loans	4723



91% of the people went for cash loans whereas the remaining 9% opted for revolving loans.

- Analysing the Gender ratio:

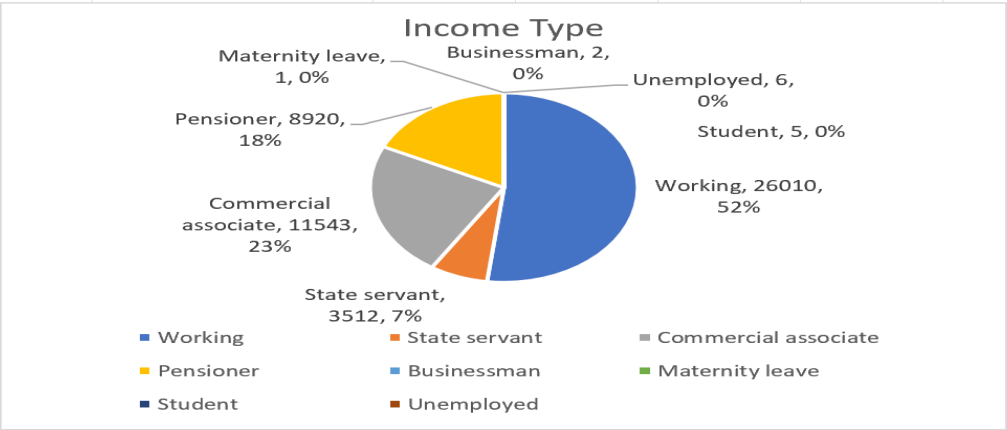
Gender	Count
M	17174
F	32825



There are 66% females and 34% males present in the dataset.

- Analysing the Income Type,

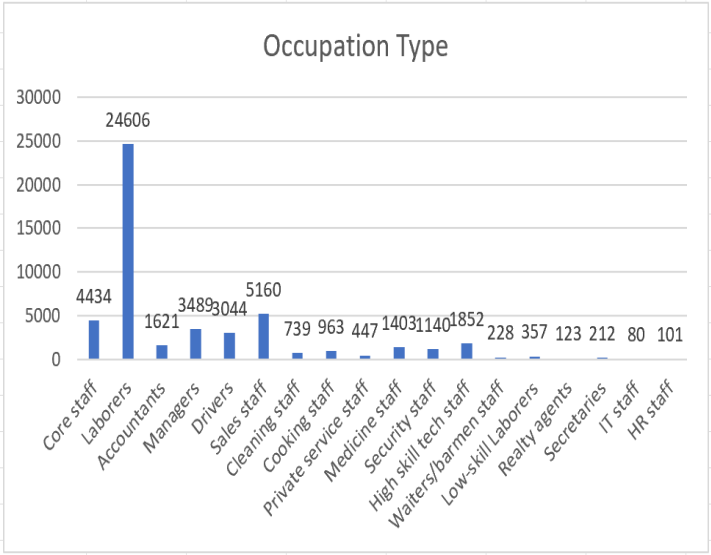
Type	Count
Working	26010
State servant	3512
Commercial associate	11543
Pensioner	8920
Businessman	2
Maternity leave	1
Student	5
Unemployed	6



Working had the highest percentage followed by Commercial and Pensioner. Categories mentioned as 0% were a very small part of the population, thus, they can be considered irrelevant.

- Analysing occupation types:

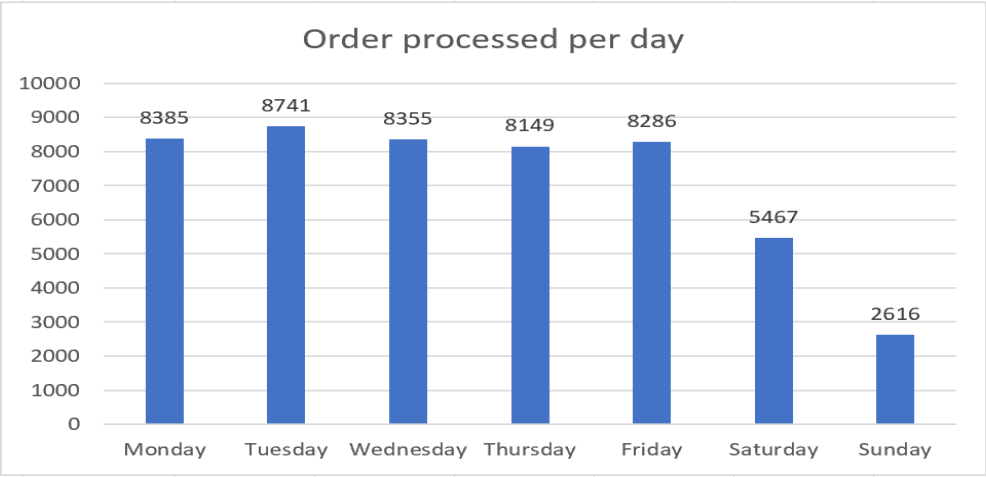
Type	Count
Core staff	4434
Laborers	24606
Accountants	1621
Managers	3489
Drivers	3044
Sales staff	5160
Cleaning staff	739
Cooking staff	963
Private service staff	447
Medicine staff	1403
Security staff	1140
High skill tech staff	1852
Waiters/barmen staff	228
Low-skill Laborers	357
Realty agents	123
Secretaries	212
IT staff	80
HR staff	101



It can be seen from the chart that Laborers have the highest count followed by sales staff and core staff. But the Laborers are dominating the overall count by a large figure.

- Analysing the number of orders processed each day of the week:

Day	Count
Monday	8385
Tuesday	8741
Wednesday	8355
Thursday	8149
Friday	8286
Saturday	5467
Sunday	2616

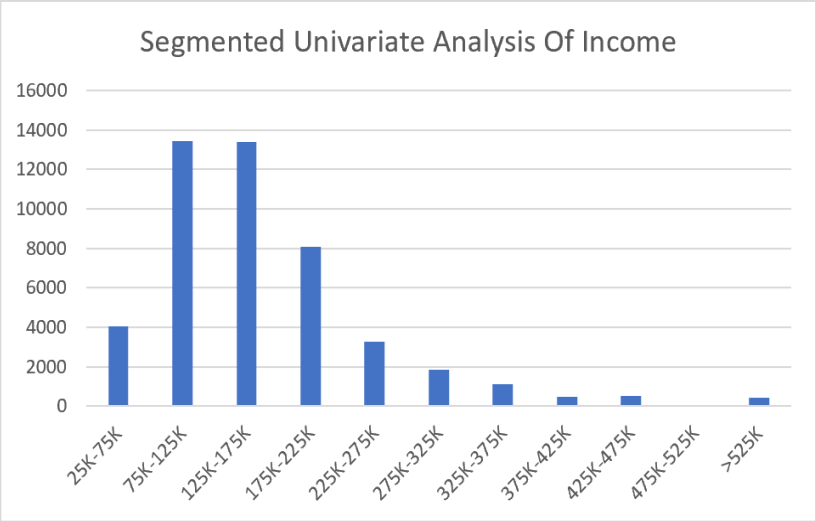


Each day has nearly the same number of orders processed, the count lowers at Saturday and Sunday. This is probably due to these days being weekends.

Task D: Perform univariate analysis to understand the distribution of individual variables, segmented univariate analysis to compare variable distributions for different scenarios, and bivariate analysis to explore relationships between variables and the target variable using Excel functions and features.

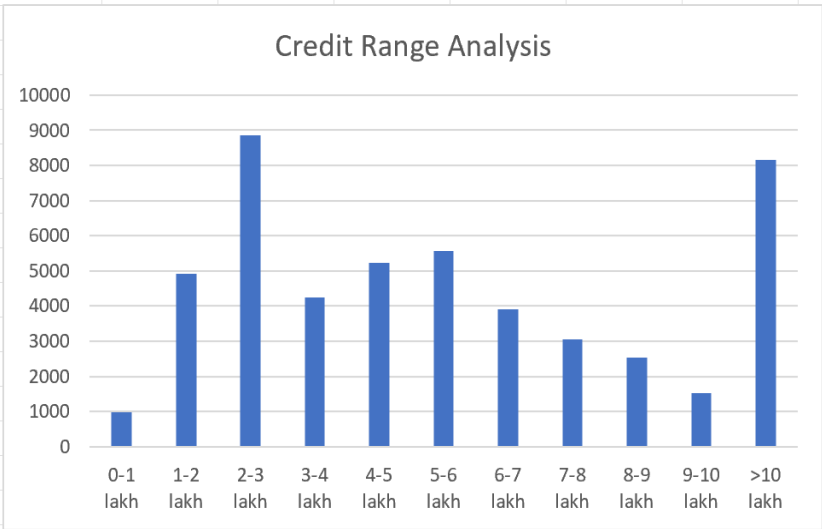
- Univariate analysis has been shown in Task C for different columns, they also helped analyze if there was data imbalance.
- Segmented Univariate Analysis of Income:

Income Class	Count
25K-75K	4030
75K-125K	13410
125K-175K	13365
175K-225K	8082
225K-275K	3268
275K-325K	1861
325K-375K	1103
375K-425K	489
425K-475K	513
475K-525K	56
>525K	445



- The greatest number of people can be found in the 75K-125K followed almost equally by 125K-175K. The least number of people can be found in 475K-525K range.

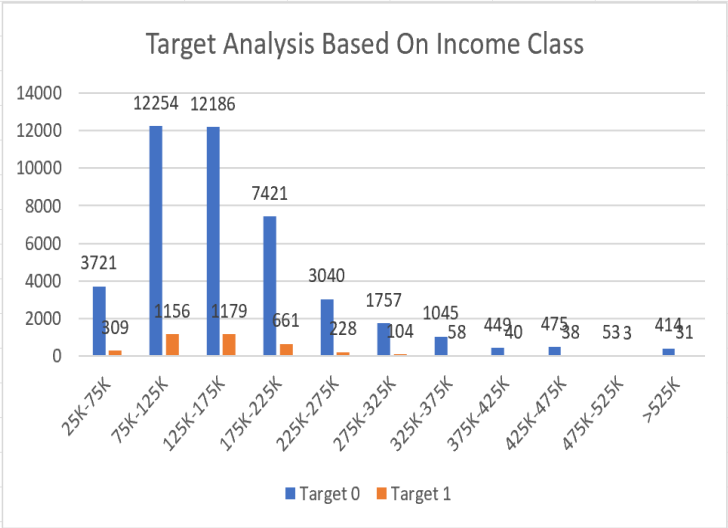
Credit Range	Count
0-1 lakh	989
1-2 lakh	4911
2-3 lakh	8849
3-4 lakh	4256
4-5 lakh	5228
5-6 lakh	5554
6-7 lakh	3909
7-8 lakh	3062
8-9 lakh	2547
9-10 lakh	1524
>10 lakh	8146



- The greatest number of people lie in the 2-3 lakh range followed by >10 lakh range whereas the least number of people can be found in the 0-1 lakh range followed closely by 9-10 lakh range.

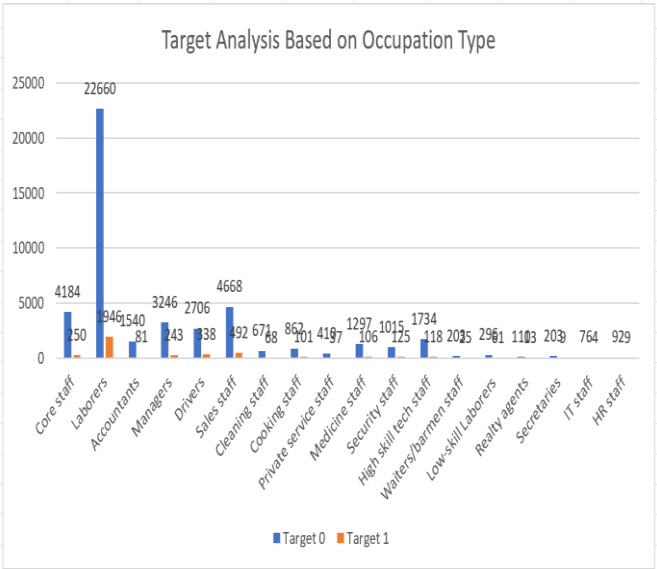
- Bivariate analysis will be performed here for multiple columns.
  - Analysing the targets based on income class –

Income Class	Target 0	Target 1
25K-75K	3721	309
75K-125K	12254	1156
125K-175K	12186	1179
175K-225K	7421	661
225K-275K	3040	228
275K-325K	1757	104
325K-375K	1045	58
375K-425K	449	40
425K-475K	475	38
475K-525K	53	3
>525K	414	31



  - Analysing targets based on occupation type –

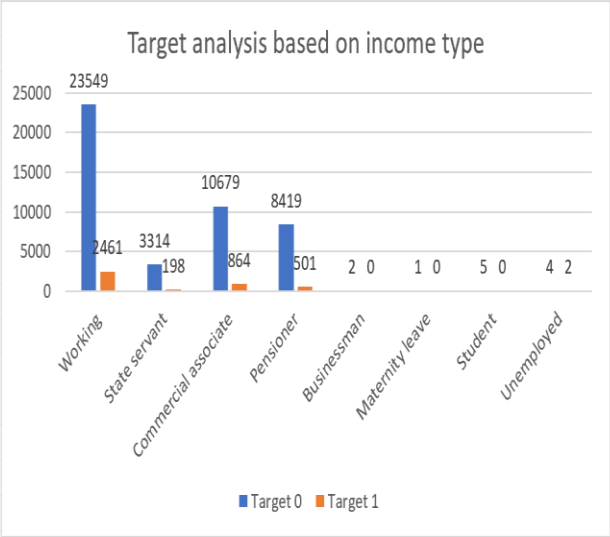
Type	Target 0	Target 1	Total	Defaulters
Core staff	4184	250	4434	5.64%
Laborers	22660	1946	24606	7.91%
Accountants	1540	81	1621	5.00%
Managers	3246	243	3489	6.96%
Drivers	2706	338	3044	11.10%
Sales staff	4668	492	5160	9.53%
Cleaning staff	671	68	739	9.20%
Cooking staff	862	101	963	10.49%
Private service staff	410	37	447	8.28%
Medicine staff	1297	106	1403	7.56%
Security staff	1015	125	1140	10.96%
High skill tech staff	1734	118	1852	6.37%
Waiters/barmen staff	203	25	228	10.96%
Low-skill Laborers	296	61	357	17.09%
Realty agents	110	13	123	10.57%
Secretaries	203	9	212	4.25%
IT staff	76	4	80	5.00%
HR staff	92	9	101	8.91%



Laborers had the greatest number of timely payments followed by sales staff. The default rate is found to be the highest for Drivers followed closely by Waiters/barmen staff.

- Target Analysis based on income type –

Type	Target 0	Target 1	Total	Defaulters
Working	23549	2461	26010	9.46%
State servant	3314	198	3512	5.64%
Commercial associate	10679	864	11543	7.49%
Pensioner	8419	501	8920	5.62%
Businessman	2	0	2	0.00%
Maternity leave	1	0	1	0.00%
Student	5	0	5	0.00%
Unemployed	4	2	6	33.33%



Pensioners had the lowest number of defaulters followed closely by State servants. Whereas, Unemployed people had the highest defaulters rate followed by Working class.

Task E: Segment the dataset based on different scenarios (e.g., clients with payment difficulties and all other cases) and identify the top correlations for each segmented data using Excel functions.

Correlations for target 0

Categories	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_ANNUITY	AMT_GOODS_PRICE	REGION_POPULATION	DAYS_BIRTH	DAYS_EMPLOYED	DAYS_REGISTRATION	DAYS_ID_PUBLISH	CNT_FAM_MEMBERS
CNT_CHILDREN	1.000	0.036	0.006	0.026	0.002	-0.025	0.336	0.028	0.183	-0.033	0.879
AMT_INCOME_TOTAL	0.036	1.000	0.378	0.451	0.385	0.182	0.074	-0.056	0.069	0.032	0.042
AMT_CREDIT	0.006	0.378	1.000	0.771	0.987	0.096	-0.051	-0.096	0.008	-0.008	0.065
AMT_ANNUITY	0.026	0.451	0.771	1.000	0.776	0.117	0.010	-0.071	0.035	0.009	0.078
AMT_GOODS_PRICE	0.002	0.385	0.987	0.776	1.000	0.099	-0.049	-0.097	0.011	-0.009	0.063
REGION_POPULATION_RELATIVE	-0.025	0.182	0.096	0.117	0.099	1.000	-0.030	0.004	-0.059	-0.002	-0.023
DAYS_BIRTH	0.336	0.074	-0.051	0.010	-0.049	-0.030	1.000	0.177	0.335	0.270	0.284
DAYS_EMPLOYED	0.028	-0.056	-0.096	-0.071	-0.097	0.004	0.177	1.000	0.116	0.036	0.000
DAYS_REGISTRATION	0.183	0.069	0.008	0.035	0.011	-0.059	0.335	0.116	1.000	0.104	0.171
DAYS_ID_PUBLISH	-0.033	0.032	-0.008	0.009	-0.009	-0.002	0.270	0.036	0.104	1.000	-0.025
CNT_FAM_MEMBERS	0.879	0.042	0.065	0.078	0.063	-0.023	0.284	0.000	0.171	-0.025	1.000

Correlations for target 1

Categories	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_ANNUITY	AMT_GOODS_PRICE	REGION_POPULATION	DAYS_BIRTH	DAYS_EMPLOYED	DAYS_REGISTRATION	DAYS_ID_PUBLISH	CNT_FAM_MEMBERS
CNT_CHILDREN	1.000	-0.003	0.008	0.029	-0.001	-0.020	0.250	0.010	0.152	-0.042	0.893
AMT_INCOME_TOTAL	-0.003	1.000	0.307	0.377	0.307	0.099	0.010	-0.024	0.030	0.001	0.006
AMT_CREDIT	0.008	0.307	1.000	0.750	0.982	0.068	-0.143	-0.097	-0.043	-0.044	0.061
AMT_ANNUITY	0.029	0.377	0.750	1.000	0.749	0.073	-0.009	-0.053	0.022	-0.021	0.076
AMT_GOODS_PRICE	-0.001	0.307	0.982	0.749	1.000	0.077	-0.141	-0.103	-0.043	-0.050	0.055
REGION_POPULATION_RELATIVE	-0.020	0.099	0.068	0.073	0.077	1.000	-0.016	0.001	-0.046	-0.005	-0.017
DAYS_BIRTH	0.250	0.010	-0.143	-0.009	-0.141	-0.016	1.000	0.222	0.288	0.248	0.199
DAYS_EMPLOYED	0.010	-0.024	-0.097	-0.053	-0.103	0.001	0.222	1.000	0.126	0.085	-0.012
DAYS_REGISTRATION	0.152	0.030	-0.043	0.022	-0.043	-0.046	0.288	0.126	1.000	0.090	0.152
DAYS_ID_PUBLISH	-0.042	0.001	-0.044	-0.021	-0.050	-0.005	0.248	0.085	0.090	1.000	-0.044
CNT_FAM_MEMBERS	0.893	0.006	0.061	0.076	0.055	-0.017	0.199	-0.012	0.152	-0.044	1.000

- The negative correlations have been highlighted as red, the top 10 correlations have been highlighted as yellow and correlation between the same columns have been highlighted as green.
- For target 0, the correlation between AMT_GOODS_PRICE and AMT_CREDIT was the highest at 0.987, indicating they move almost hand-in-hand. The lowest correlation was between DAY_EMPLOYED and AMT_GOODS_PRICE.
- For target 1, the correlation between AMT_GOODS_PRICE and AMT_CREDIT was the highest at 0.982, indicating the same behaviour as target 0. The lowest correlation was between DAYS_BIRTH and AMT_CREDIT.

Result: Through this project, I learnt a lot about the use cases of different formulas in excel. I learnt about the process of data cleaning by understanding how to handle missing value along with outliers in relation to the context of data and how those columns affect the analysis. I learnt a great deal about the different types of analysis and how they can be used to find relation amongst different data (Univariate, Segmented Univariate, Bivariate). I also learnt the art of conditional formatting.

Alongside all these important learnings, I also built a good background with the process of Bank loans and how they work. I learnt about this by researching each of the columns and understanding what each of them meant.

Excel Link:

https://docs.google.com/spreadsheets/d/1o6euTDC4LOrO1jhMaE848yJZzTJ-7SSL/edit?usp=sharing&ouid=100878477496382804997&rtpof=true&sd=true